Math Refresher for Masters

Homework 6

1 Points: $(0,2)$, $(1,3)$, $(2,8)$.
$p \in \mathcal{P}_2$

$$p(t) = y_0 L_0(t) + y_1 L_1(t) + y_2 L_2(t)$$

$$L_i(t) = \prod_{j \neq i} \frac{t - t_j}{t_i - t_j}$$

1) $t_0 = 0$: $L_0(t) = \frac{(t-1)(t-2)}{(0-1)(0-2)} = \frac{1}{2}(t-1)(t-2)$

$t_1 = 1$: $L_1(t) = \frac{t(t-2)}{(1-0)(1-2)} = -t(t-2)$

$t_2 = 2$: $L_2(t) = \frac{t(t-1)}{(2-0)(2-1)} = \frac{1}{2}t(t-1)$

2) $p(t) = (t-1)(t-2) - 3t(t-2) + 4t(t-1) =$
$= t^2 - 3t + 2 - 3t^2 + 6t + 4t^2 - 4t =$
$= \boxed{2t^2 - t + 2}$

3) $p(\frac{1}{2}) = 2 \cdot \frac{1}{4} - \frac{1}{2} + 2 = \boxed{2}$

2 $f(t) = e^t$ $\quad t_0 = 0$, $t_1 = 1$

$f(0) = e^0 = 1$
$f(1) = e^1 = e$

$L_0(t) = \frac{t-1}{0-1} = 1 - t$ $\qquad L_1(t) = \frac{t}{1-0} = t$

$p(t) = (1-t) \cdot 1 + te = 1 + (e-1)t$

$f(t) - p(t) = \frac{f''(\xi)}{2!}(t-t_0)(t-t_1)$, $\xi \in [0;1]$

$f''(t) = e^t$, if $t \in [0;1] \Rightarrow |f''(t)| \leq e \Rightarrow |f(t) - p(t)| \leq \frac{e}{2}|t(t-1)|$

$\max_{t \in [0;1]} |t(t-1)| = \left[t = \frac{1}{2}\right] = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \Rightarrow \boxed{|f(t) - p(t)| \leq \frac{e}{8}}$

if $t=\frac{1}{2}$: $f(\frac{1}{2}) = e^{\frac{1}{2}}$

$$p(\tfrac{1}{2}) = 1 + \frac{e-1}{2} = \frac{1+e}{2}$$

$$E(\tfrac{1}{2}) = \left| e^{\frac{1}{2}} - \frac{1+e}{2} \right| \approx \boxed{0.21}$$

3 $f(t) = t^3$ on $[-1;1]$ $\quad V = \{1, t\}$

$$\langle u, v \rangle = \int_{-1}^{1} u(t) \cdot v(t)\,dt \qquad p(t) = a + bt$$

1) $\begin{cases} \langle f-p, 1 \rangle = 0 \\ \langle f-p, t \rangle = 0 \end{cases}$

$$\langle t^3 - a - bt, 1 \rangle = \langle t^3, 1 \rangle - a\langle 1,1 \rangle - b\langle t, 1 \rangle = 0$$

$$\Rightarrow \begin{pmatrix} \langle 1,1 \rangle & \langle t,1 \rangle \\ \langle 1,t \rangle & \langle t,t \rangle \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \langle t^3, 1 \rangle \\ \langle t^3, t \rangle \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 \\ 0 & \frac{2}{3} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{2}{5} \end{pmatrix} \quad \Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{3}{5} \end{pmatrix}$$

$$\boxed{p(t) = \frac{3}{5}t}$$

Check: $r(t) = f(t) - p(t) = t^3 - \frac{3}{5}t$

$$\langle r, 1 \rangle = \int_{-1}^{1} \left(t^3 - \frac{3}{5}t\right) dt = \left(\frac{t^4}{4} - \frac{3t^2}{10}\right)\Big|_{-1}^{1} = 0$$

$$\langle r, t \rangle = \int_{-1}^{1} \left(t^3 - \frac{3}{5}t\right) t\,dt = \left(\frac{t^5}{5} - \frac{t^3}{5}\right)\Big|_{-1}^{1} = \frac{2}{5} - \frac{2}{5} = 0 \quad \Rightarrow \text{true}$$

4 Points: $(-1,0), (0,1), (1,1), (2,2)$ $\qquad p(t) = c_0 + c_1 t$

$$B = \begin{pmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} \qquad y = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 2 \end{pmatrix} \qquad c = \begin{pmatrix} c_0 \\ c_1 \end{pmatrix}$$

$$B^T Bc = B^T y$$

$$B^T B = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 6 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 2 \\ 2 & 6 \end{pmatrix} c = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$

$$c_0 = 1 - \frac{c_1}{2}$$

$$c_0 = \frac{5}{2} - 3c_1$$

$$\Rightarrow 1 - \frac{c_1}{2} = \frac{5}{2} - 3c_1$$

$$2 - c_1 = 5 - 6c_1$$

$$5c_1 = 3$$

$$c_1 = \frac{3}{5} \quad \Rightarrow c_0 = 1 - \frac{3}{10} = \frac{7}{10}$$

$$\boxed{p(t) = \frac{7}{10} + \frac{3}{5}t}$$

$$r = y - Bc = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} \frac{7}{10} \\ \frac{13}{10} \\ \frac{19}{10} \\ \frac{25}{10} \end{pmatrix} = \begin{pmatrix} -\frac{7}{10} \\ -\frac{3}{10} \\ \frac{1}{10} \\ \frac{5}{10} \end{pmatrix}$$

$$B^T r = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} -\frac{7}{10} \\ -\frac{3}{10} \\ \frac{1}{10} \\ \frac{5}{10} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\|r\|_2 = \sqrt{\frac{1}{5}}$$

5 $g(t) = c_0 + c_1 t + c_2 \sin t$

$$B = \begin{pmatrix} 1 & -1 & -\sin 1 \\ 1 & 0 & 0 \\ 1 & 1 & \sin 1 \\ 1 & 2 & \sin 2 \end{pmatrix} \qquad y = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 2 \end{pmatrix}$$

Calculate with the code: $c = \begin{pmatrix} 0.67 \\ 0.86 \\ -0.43 \end{pmatrix} \Rightarrow g(t) = 0.67 + 0.86t - 0.43\sin t$

$$r = \begin{pmatrix} -\frac{1}{6} \\ \frac{1}{3} \\ -\frac{1}{6} \\ 0 \end{pmatrix} \qquad \|r\|_2 \approx 0.408$$

$$B^T r = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$